

CM-Galaxy

A Sub-Quadratic, Attention-Free Backbone for Cheap Long-Context Modeling

A technical teaser — concept, measured results, and honest limits

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Version v0 (teaser) · June 2026 · Licensed under CC BY 4.0 · DOI: [10.5281/zenodo.21104122](https://doi.org/10.5281/zenodo.21104122) (Zenodo)

Abstract

We present **CM-Galaxy**, a small ($\approx 25\text{M}$ -parameter), attention-free, sub-quadratic language-model backbone aimed at cheap long-context modeling—sub-quadratic in both training and inference. In place of attention’s token-by-token lookup, it folds the sequence into a constant-size *associative memory* together with a convolutional (long-convolution / FFT) mixer, so the per-token cost of decoding stays flat as context grows. We characterize the model honestly at small scale rather than claiming frontier quality. On the independent Zoology MQAR associative-recall benchmark the memory matches attention ($\approx 99.5\text{--}99.9\%$ across sequence lengths 64–512, over three seeds); trained at 8k context on WikiText-103, the 25M model shows healthy language modeling (≈ 37.5 perplexity, $\approx 18.1\%$ LAMBADA). Its efficiency advantage is structural: decode memory stays flat at $\approx 312\text{MB}$ from 8k to 8.4M tokens—about $30\times$ less than an attention key/value cache at 1M tokens, and $\approx 235\times$ less at 8.4M—and at high batch it sustains roughly $92\times\text{--}159\times$ more tokens per second per GPU, where the attention baseline runs out of memory and CM-Galaxy does not. We are equally explicit about the limits: long-range recall holds up better than the attention baseline yet stays modest in absolute terms ($\approx 23\%$ vs $\approx 16\%$ at a 16k gap); recall is characterized to 16k as extrapolation; the memory and throughput measurements use random weights, so they characterize speed and memory, not output quality; single-stream latency favors attention; and no comparison with state-space models (Mamba, RWKV) or larger scales has been run. We release this as a teaser—concept, measured results, and honest limits—and outline an evidence-driven path, a controlled paged memory, toward the open problem of raising long-context recall.

1 Introduction

Modern language models are built on attention [1], which lets every token attend to every other token. This is expressive, but expensive: the cost of generating each new token grows with the length of the context. During autoregressive decoding the model maintains a key/value (KV) cache whose size grows linearly with the number of tokens seen, and the attention computation itself grows quadratically with sequence length. For long-context inference—long documents, long conversations, large retrieved contexts—this cost, in both memory and time, becomes the dominant constraint on what is affordable to serve.

Most widely used responses to this constraint work *around* attention rather than changing it. Retrieval-augmented generation [2] attaches an external retriever to an otherwise unchanged

attention model: it reduces how many tokens must be fed in, but it does not change attention’s per-token cost. Mixture-of-experts [3, 4] sparsifies the feed-forward layers, lowering compute per token, while the attention KV cache still grows with context. These are valuable techniques; they simply do not remove the underlying growth of attention’s state. A separate line of work attacks the cost at the architectural level, replacing attention with mechanisms whose decoding state does not grow with context: state-space and linear-recurrent models such as Mamba [5] and RWKV [6], long-convolution mixers such as Hyena [7], and delta-rule memories such as DeltaNet [8] and Gated DeltaNet [9]. These sub-quadratic models trade some of attention’s raw power for a constant- or slow-growing state, and so for far cheaper long-context inference.

CM-Galaxy belongs to this attention-free, sub-quadratic family. It is a small (≈ 25 M-parameter) backbone that combines a sub-quadratic token mixer (in the long-convolution / FFT family) with a gated delta-rule associative memory (in the Gated DeltaNet family). Its defining choice is a constant-size decode state: rather than re-attending to a growing set of past tokens, it carries forward a fixed, compressed representation of the context, so the per-token cost of decoding stays flat as context grows. The building blocks are therefore drawn from known families; the contribution of this work is not a new primitive but the specific attention-free combination and, above all, an honest and reproducible characterization of what such a small model does and does not achieve.

This document is a teaser, and we are deliberate about the line between what we claim and what we withhold. We claim only measured results: recall on par with attention on a standard associative-recall benchmark, healthy language modeling at 8k context, and a structural efficiency advantage in decode memory and batched throughput—each reported together with its limits, including a long-context recall gap that we do not paper over. We withhold, for now, the architecture recipe, the training procedure, the model source code, and the trained checkpoints; the standard-transformer baseline used for the efficiency comparison can be shared separately so that those measurements may be independently reproduced. The work was carried out independently, and every component retained in the model earned its place through controlled ablation.

2 The CM-Galaxy approach

CM-Galaxy (CM for *Continuous Medium*¹) is an attention-free, sub-quadratic backbone assembled from two complementary parts. The first is a *sub-quadratic token mixer* in the long-convolution / FFT family [7], which captures local and sequential structure in $O(n \log n)$ time without comparing tokens pairwise. The second is a *gated delta-rule associative memory* in the Gated DeltaNet family [8, 9]: a content-addressed key–value store written and read by a gated delta update, giving the model an explicit mechanism for recalling earlier content.

The two parts play different roles, and the design rests on keeping them separate: the mixer is responsible for cheap local mixing, the associative memory for content-addressed recall. Neither maintains a state that grows with context. Instead of comparing each new token against a growing set of past tokens—attention’s token-by-token lookup—the model folds

¹*Continuous Medium* (CM) is a physics framework previously developed by the author (26+ papers on Zenodo; ORCID 0009-0003-7176-2395), from which several of CM-Galaxy’s design choices were initially drawn. The physics motivated the *search space*, not the claims: each component is kept only where controlled ablation earns it, and several physics-motivated modules were tested and dropped. All results in this paper rest on standard benchmarks and ablations.

the sequence into a fixed-size, compressed representation and carries it forward. The decode state is therefore constant in size, and the cost of generating each additional token stays flat as the context lengthens; this is the property that the efficiency results in Section 3 quantify.

Why combine the two parts rather than rely on the mixer alone? Because a sub-quadratic mixer, by itself, is a strong local pattern matcher but a poor content-addressed retriever—a gap analyzed for gated-convolution models by Arora et al. [10], who introduce the multi-query associative-recall (MQAR) benchmark for exactly this purpose. We observe the same effect directly: a pure-mixer variant at roughly 70M parameters reaches strong language-modeling perplexity (about 30 on WikiText-103) yet collapses on associative recall, scoring around 14% on MQAR against about 71% for an attention baseline, and under 1% on LAMBADA. The failure is one of recall, not of language modeling—and it is precisely what motivated adding a dedicated delta-rule memory. The memory block is what restores recall to the level reported in Section 3, while the mixer keeps the model as a whole sub-quadratic.

What we describe here is the *concept* and the *component families*, not the recipe. The specific mixer parameterization, the memory’s internal configuration, the number and placement of layers, the training curriculum, the source code, and the trained checkpoints are withheld at this stage (Section 7). Only the standard-transformer baseline used for the efficiency comparison is offered for independent verification.

3 What is demonstrated

The results below come from a single small (≈ 25 M-parameter) model and a standard attention baseline, measured directly. They are small-scale figures—the contribution is the architecture and its efficiency, not absolute quality—but they are real and mutually consistent.

3.1 Associative recall matches attention

On MQAR [10], the memory block recalls about as well as attention. Table 1 reports accuracy across sequence lengths from 64 to 512 in the independent Zoology harness. Across the ladder the memory stays within roughly half a percentage point of attention, and both models learn the task within a few epochs. Notably, the memory solves MQAR *without* the short-convolution “crutch” that the attention baseline relies on in this harness, which indicates that the recall comes from the memory mechanism itself rather than from local convolution. At the longest length tested the memory trails attention slightly, and with more run-to-run variance—very close to parity, but not identical.

Sequence length	Memory (delta-rule)	Attention
64	$\approx 99.9\%$	$\approx 100\%$
128	$\approx 99.7\%$	$\approx 100\%$
256	$\approx 99.8\%$	$\approx 100\%$
512	$\approx 99.5\%$	$\approx 100\%$

Table 1: MQAR (multi-query associative recall) accuracy in the independent Zoology harness. Results at 256 and 512 are averaged over three seeds (e.g. 99.1–99.8% at 512); both models reach these levels within a few epochs. The number of key–value pairs grows with sequence length (16 at 128, up to 64 at 512), so the task becomes harder along the ladder.

3.2 Recall over long gaps

Clean, short-range recall is one thing; holding a memory across a long gap is harder. On a real-token induction task—predicting a token that reappears after a long filler gap—we compare CM-Galaxy against the same attention baseline in one shared harness, both trained to ≤ 2048 context (so the longer gaps are extrapolation). Table 2 reports accuracy as the gap grows to 16k, with both models evaluated under memory-efficient (FlashAttention) attention so that neither runs out of memory. CM-Galaxy’s recall stays roughly flat with distance, while the attention baseline’s declines steadily at the longer gaps. From about 4k onward CM-Galaxy’s recall is the higher of the two, and the margin widens with distance—at a 16k gap CM-Galaxy holds $\approx 23\%$ against the attention baseline’s $\approx 16\%$. Two honest qualifications: the absolute numbers are modest at this scale, and the attention baseline carries roughly 30% more parameters ($\approx 32\text{M}$ vs $\approx 25\text{M}$). Perplexity stays stable across gaps, and a separate run training CM-Galaxy *directly* at 8k reproduces the same flat recall—so this reflects genuine memory capacity, not an extrapolation artefact.

Gap (tokens)	CM-Galaxy	Attention
0	38.6	33.6
512	29.6	26.6
2048	25.9	25.4
4096	23.6	21.1
8192	24.0	18.4
16384	22.7	16.4

Table 2: Long-gap induction recall (%) on real WikiText-103 tokens in one shared harness; both models use memory-efficient (FlashAttention) attention and are trained to ≤ 2048 context, so the longer gaps are extrapolation. CM-Galaxy’s recall stays roughly flat as the gap grows, whereas the attention baseline declines steadily. Absolute values are modest at this scale, and the attention baseline carries $\approx 30\%$ more parameters. A separate run training CM-Galaxy directly at 8k reproduces this flat recall, indicating genuine capacity rather than extrapolation drift.

3.3 Language modeling at 8k

Trained directly at 8k context on WikiText-103 [11], the 25M model reaches ≈ 37.5 perplexity with $\approx 18.1\%$ LAMBADA last-word accuracy [12]. These are modest numbers in absolute terms, as expected at this scale, but well above chance and consistent with genuine language learning. Recall over long gaps *within* this 8k setting is a separate, harder question, taken up under Honest limits.

3.4 Flat decode memory

The defining efficiency property is that the decode state does not grow with context. Figure 1 shows peak decode memory as the sequence grows from 8k to 8.4M tokens: CM-Galaxy stays flat at $\approx 312\text{ MB}$, while a standard attention KV-cache grows linearly into the tens of gigabytes.

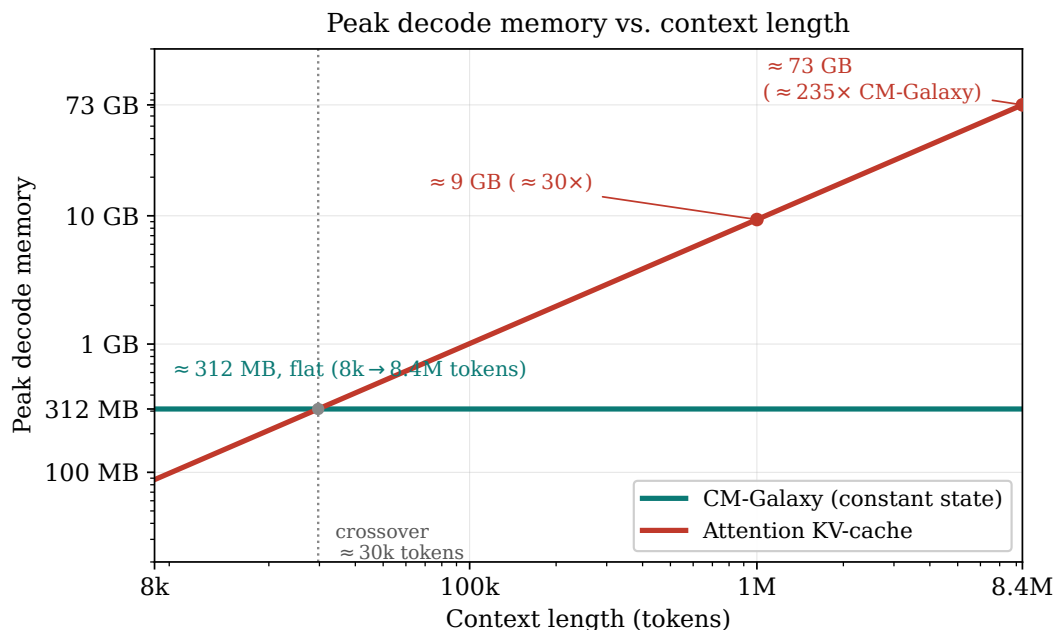


Figure 1: Peak decode memory vs. context length (log–log). CM-Galaxy’s decode state is constant, so its peak memory stays flat at ≈ 312 MB from 8k to 8.4M tokens; a standard attention KV-cache grows linearly, reaching ≈ 73 GB at 8.4M tokens—about $30\times$ more than CM-Galaxy at 1M tokens and $\approx 235\times$ more at 8.4M. Below ≈ 30 k tokens the attention cache is smaller, since CM-Galaxy carries a fixed overhead; beyond that crossover, its flat memory wins by a widening margin. Measured with randomly-initialized weights (memory only).

3.5 Throughput at the memory limit

Flat memory turns into throughput when many sequences are served at once. At high batch sizes CM-Galaxy sustains roughly $92\times$ (at 8k context) and $159\times$ (at 16k) more tokens per second per GPU than the attention baseline; the baseline runs out of memory at a batch size of about 512–1024, whereas CM-Galaxy reaches 8192 with memory still flat. Single-stream (batch-one) latency is a different story, taken up under Honest limits.

3.6 How it was measured

The speed and memory figures were collected on a single NVIDIA A100-80GB, with CM-Galaxy and the baseline run together in the same session, so the *ratios* are a fair same-hardware comparison (absolute tokens-per-second are hardware-specific and are not the claim). The baseline is a standard transformer using scaled dot-product (FlashAttention-style) attention [13], not a weakened one. These speed and memory measurements use randomly-initialized weights: they characterize cost only, not output quality. The out-of-memory thresholds quoted above are specific to the 80 GB card; on smaller GPUs attention reaches them sooner.

3.7 The overall picture

Taken together, the results form a consistent picture. The recall, perplexity and LAMBADA numbers show that the model genuinely *works*—it learns language and does clean recall on

par with attention. The memory and throughput numbers show that it *survives at scale*—its cost stays flat exactly where attention’s cost explodes. The efficiency advantage is real and grows with context and batch. What the model does *not* yet show is high recall in *absolute* terms at long range: its long-gap recall holds up better than the attention baseline’s, yet stays modest at this scale—which the next section takes up.

4 Honest limits

The results above are bounded by limits that we state as plainly as the results themselves.

The most important is the absolute level of long-range recall. Clean, short-range recall matches attention (Section 3.1), and on the long-gap induction task CM-Galaxy actually holds up *better* than the attention baseline as the gap grows (Section 3.2)—but in absolute terms the numbers are modest, around 23–24% at the longest gaps (out to 16k). Recall degrades gracefully with distance rather than collapsing, and the dominant failure mode is interference between competing items rather than a failure to store them; still, 24–27% is far short of the near-perfect recall one would ultimately want, and closing that absolute gap is the open problem. This recall has been evaluated out to 16k and confirmed as genuine capacity by training CM-Galaxy directly at 8k; 32k, 128k, and 1M remain untested. Memory stays flat out to 8.4M tokens, but whether recall stays *useful* at very long context is open.

Two further caveats concern what the efficiency numbers do and do not establish. First, the memory and throughput measurements use randomly-initialized weights, so they characterize speed and memory only—they say nothing about output quality at those lengths. Second, the efficiency advantage is specifically a batched-serving one: at single-stream (batch-one) latency the attention baseline is faster per token, by roughly 1.4–1.5 $\times$. CM-Galaxy’s win is total throughput at the memory limit, not latency for a single sequence.

Finally, the scope of what has been tested is narrow. The model has been studied at ≈ 25 M parameters; larger scales (70M, 125M, and beyond) have not been trained, so nothing here speaks to how the architecture behaves when scaled. No controlled comparison has been run against other sub-quadratic families such as Mamba [5] or RWKV [6], so CM-Galaxy should not be read as outperforming them—it is characterized only on its own terms. And no Indic-language modeling has been done; that intended application remains future work.

5 Where it goes next

The limits above define the work that remains; none of it is solved here, and most of it will need collaboration and compute.

Raising long-context recall. The central open problem is lifting recall at long range and extending it well past 8k without surrendering the efficiency. The flat-memory result is what makes the most natural path *affordable*: because the decode state is so cheap (Figure 1), there is room to let it grow in a *controlled, sub-linear* way—a chunked or paged memory that holds more of the context while staying far below an attention KV-cache. This idea is being actively explored; Memory Caching [14], for example, caches compressed segment states so that effective memory grows with length, interpolating between the fixed memory of recurrent models and the growing memory of Transformers. CM-Galaxy’s flat memory makes such growth affordable, but a genuine contribution here would have to go beyond simply growing the state—being selective about what is retained, or keeping decode cost

from growing in lockstep with stored memory. Whether a paged memory actually *maintains* recall at long context, at acceptable cost, is the central hypothesis still to be tested.

Indic long-context inference. This is the intended application, to be pursued once the architecture is matured and scaled. The natural first step is tokenization suited to Indic morphology; the efficiency demonstrated here would matter most for cheap, long-context inference in Indian languages. No Indic modeling has yet been done.

Scale and comparison. Validating the architecture at 70M, 125M, and beyond, and benchmarking it directly against other sub-quadratic families—Mamba [5], RWKV [6], and gated delta-rule models [9]—would establish where it actually stands; at present it is characterized only on its own terms.

Agentic and parallel-rollout workloads (speculative). The batched-throughput advantage is a natural fit for settings that run many parallel, long-context rollouts at once—agentic loops and reinforcement-learning training among them. This is untested, and such workloads also lean on the long-context recall that remains the open problem, so it is a direction to explore rather than a claim.

These are open research problems, set out plainly so that what is demonstrated today and what remains to be shown are never confused. The honest claim is narrow and real: a working, well-characterized sub-quadratic backbone with a measured efficiency advantage, and an evidence-driven path for the parts that remain open.

6 Positioning and relation to prior work

The components of CM-Galaxy are not new. Its token mixer belongs to the long-convolution / FFT family exemplified by Hyena [7]; its memory belongs to the gated delta-rule family of DeltaNet [8] and Gated DeltaNet [9]; and its constant-size recurrent state is shared with state-space and linear-recurrent models such as Mamba [5] and RWKV [6]. The contribution is not a new primitive but the specific attention-free *combination* of these parts, with a careful, reproducible *characterization* of it at small scale. The central idea is qualitative as much as quantitative: instead of matching every new token against every past one, the associative memory and the mixer summarize the context *as a whole*, and on clean associative recall this reaches essentially attention’s accuracy from a *constant-size* state—the very property that keeps memory flat at long context (Figure 1). Attention-level clean recall without a growing cache is the result the rest of the work builds on.

We are equally clear about what is *not* yet established. On long-gap recall CM-Galaxy already holds up better than the attention baseline (Section 3.2), but in *absolute* terms that recall is still modest, and raising it without surrendering the efficiency is the central open problem (Sections 4–5); recall has been evaluated to 16k. The architecture has likewise not been scaled, and no controlled comparison has been run against Mamba, RWKV, or other sub-quadratic families. CM-Galaxy is therefore positioned neither as a transformer replacement nor as a competitor to any particular sub-quadratic model, but as one carefully measured point in the attention-free design space whose standing relative to its neighbours is genuinely still to be settled.

Two points guard against an over-narrow reading. First, the efficiency is not confined to inference: because the architecture is sub-quadratic ($O(n \log n)$) rather than quadratic, its cost advantage at long context applies to *training* as well as serving, and the model is trained *directly* at 8k context, via a length curriculum, rather than merely run there—long-

context behaviour is learned, not bolted on afterwards. Second, the efficiency is not tied to any one language or task: cheap long-context modeling is useful wherever contexts are long, from documents and retrieved passages to code and multi-turn dialogue. Among these, indigenous Indic long-context modeling stands out as a particularly *underserved* opportunity—prominent Indic efforts to date are built largely on the transformer, often with mixture-of-experts [3, 4], which lowers per-token compute but leaves attention’s growing state in place—but it is one opportunity among several, not the sole target. The direction is also orthogonal to mixture-of-experts: MoE sparsifies the feed-forward computation while CM-Galaxy replaces the token mixer, so the two are complementary rather than competing.

7 Disclosure, collaboration, and citation

This document is a teaser (v0). It shares the *concept*, the *measured results*, and the *honest limits*—what CM-Galaxy does, not how it is built. The architecture recipe (the specific construction of the mixer and the memory block), the training-curriculum details, the model source code, and the trained checkpoints are not part of this release. What is shared is enough to understand the claim and probe its boundaries; what is held back is the constructive recipe.

So that the comparisons can be checked against a standard, non-weakened baseline rather than a strawman, the attention baseline used throughout is released separately as a standalone script—a plain decoder-only transformer (token embedding, rotary positions, memory-efficient scaled dot-product / FlashAttention attention, RMSNorm, SwiGLU, and a KV-cache). It contains none of CM-Galaxy’s own architecture and uses only standard, openly-used components, so anyone can reproduce a comparable baseline and confirm that the comparison is fair.

This work was designed, built, and validated independently, on rented cloud GPUs. Research collaboration is welcome on a *non-commercial* basis—in particular, validation at larger scale, controlled comparison against state-space and linear-attention families (Mamba, RWKV, gated delta-rule models), and Indic long-context pilots. Enquiries: speedgaplabs@gmail.com.

Availability. Repository (teaser, baseline script, and this paper):

<https://github.com/singhmandy25-gif/cm-galaxy>

Teaser directory: <https://github.com/singhmandy25-gif/cm-galaxy/tree/main/teaser>

DOI: <https://doi.org/10.5281/zenodo.21104122>

Cite as. Mandeep Singh (2026). *CM-Galaxy: A Sub-Quadratic, Attention-Free Backbone for Cheap Long-Context Modeling* (Teaser v0). Zenodo. DOI: 10.5281/zenodo.21104122. ORCID: 0009-0003-7176-2395.

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